

Reminders

Upcoming due dates

Wed through FRI Quiz 8 online

Wed Guest lecture

Fri A4

NEXT Wed (finals week)

Final project*, video*, team eval survey,
post-course survey

How to be wrong IV

Data Science in Practice

Jason G. Fleischer, PhD

Dept. of Cognitive Science

UC San Diego

Week 10 / Finals Week Extra Credit Opportunities

- Finish strong with your group progress surveys!
 - 0.5% EC if you fill out all 7
- Fill out the teaching evaluations (SET used to be CAPE) [Evaluations site](#)
 - Must complete evaluations BEFORE 8AM on Saturday after Week 10. No exceptions.
 - 0.5% EC if >75% of the class fills this out (only happens half the time)
- Post course survey
 - 0.25% EC

Final Projects

- **Report (15% of grade - GROUP)**
 - Can copy + paste from proposal & checkpoints & can update/edit/change
 - Adding on:
 - Choice if we have your permission to make public
 - Overview
 - Analysis (w/ “strong” data visualizations)
 - Conclusion & Discussion
 - It’s a report, not a vomiting of data.
 - Write in the diamond structure of technical reports
 - Do not have to include deadends, massive codebases, or irrelevant explorations. Can put those in a separate file that is referenced in report.
 - Do edit for correctness & conciseness at the end
- **Video (3% of grade - GROUP)**
 - 3-5 minutes
 - Post video so that it is PUBLICLY VISIBLE on Google Drive, Dropbox, TikTok, YouTube, whatever.
 - Place a link to the video right after the title of your final project notebook
 - Everyone should participate; not everyone has to be IN the video
- **Team Evaluation Survey (1% of grade - INDIVIDUAL)**
 - Give us more information about who contributed / did not contribute; this will effect individual grades
 - Will include questions that help us with the pedagogical experiment of group dynamics running this year!

Seeking PLAs for next quarter

- Holding office hours / answering student questions
- Helping write / test new assignments
- Collecting class demos, notes, videos and organizing into an online textbook
- Assisting with project feedback

Guest lecture Wednesday June 3 @ 2PM ONLY

NO MORNING LECTURE? Or do you want to do this with me only?

- Discussion (including you!): How AI will affect student's careers and society as a whole.
- Discussants:
 - Emilia Rosselli Del Turco - Hollan lab - Relationship between knowledge workers and their tools
 - Sam Taylor - Bergen lab - LLMs, how they use language, and how they deceive
 - Jack Terwilliger - Rossano lab - Social cognition in humans, animals, and LLMs
- AI real world capabilities quirks and failures including its predilection toward sycophancy and hallucination, AI impacts on economics in general, how companies and investors see AI vs what AI actually is doing, impacts on economic inequality, environmental impacts, ethical concerns including existential threat and AI consciousness, and practical concerns around the usage of AI by an undergraduate looking to get a job.
- READ THIS FIRST:
 - <https://hbr.org/2025/09/ai-generated-workslop-is-destroying-productivity>
 - <https://www.microsoft.com/en-us/worklab/work-trend-index/agents-human-agency-and-the-opportunity-for-every-organization>
 - <https://drive.google.com/file/d/1-lxpoqc6Dc0bildYa8hitmDLhnqLUPjp/view?usp=sharing>



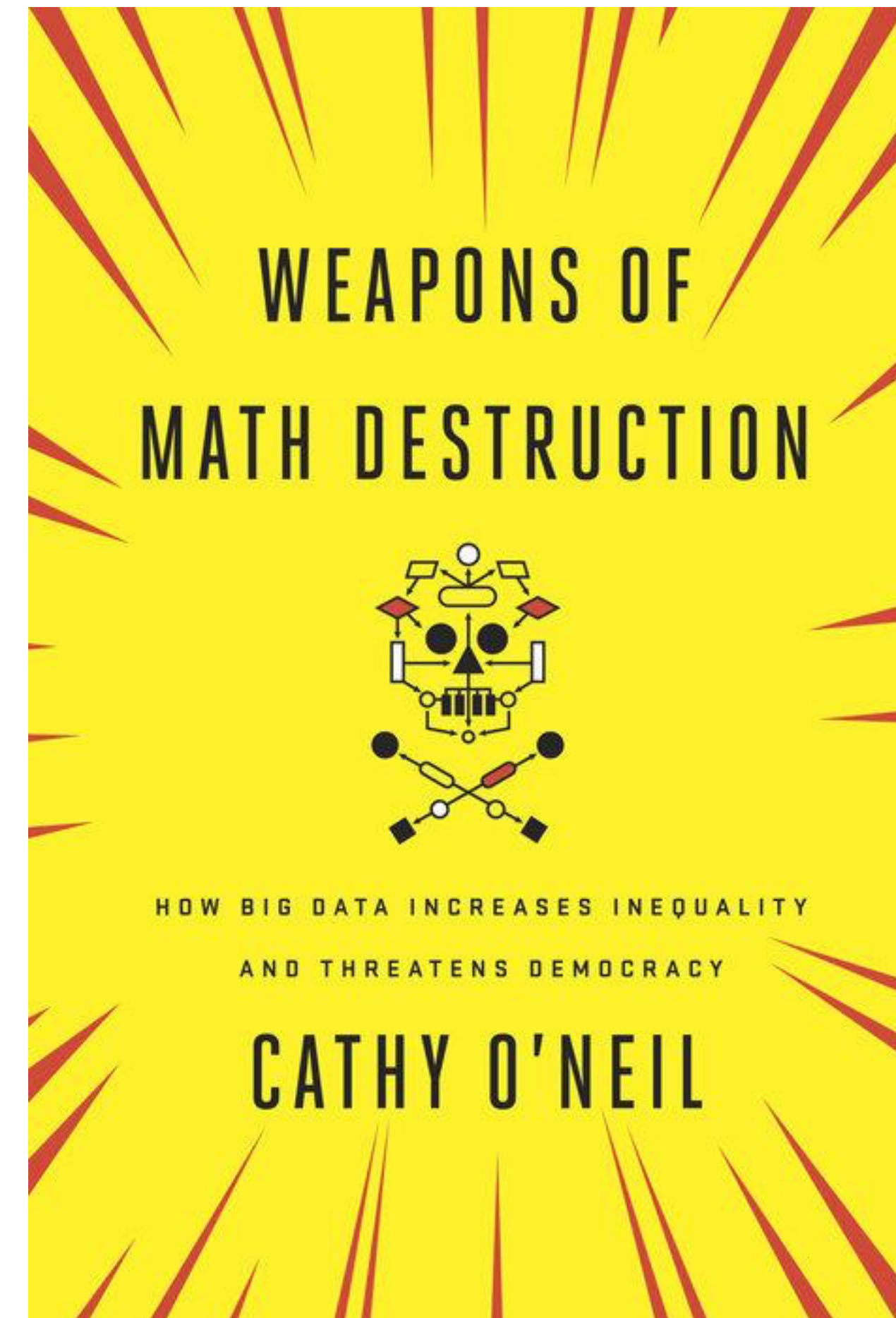
Ethical errors

Problems arising from data science and algorithms

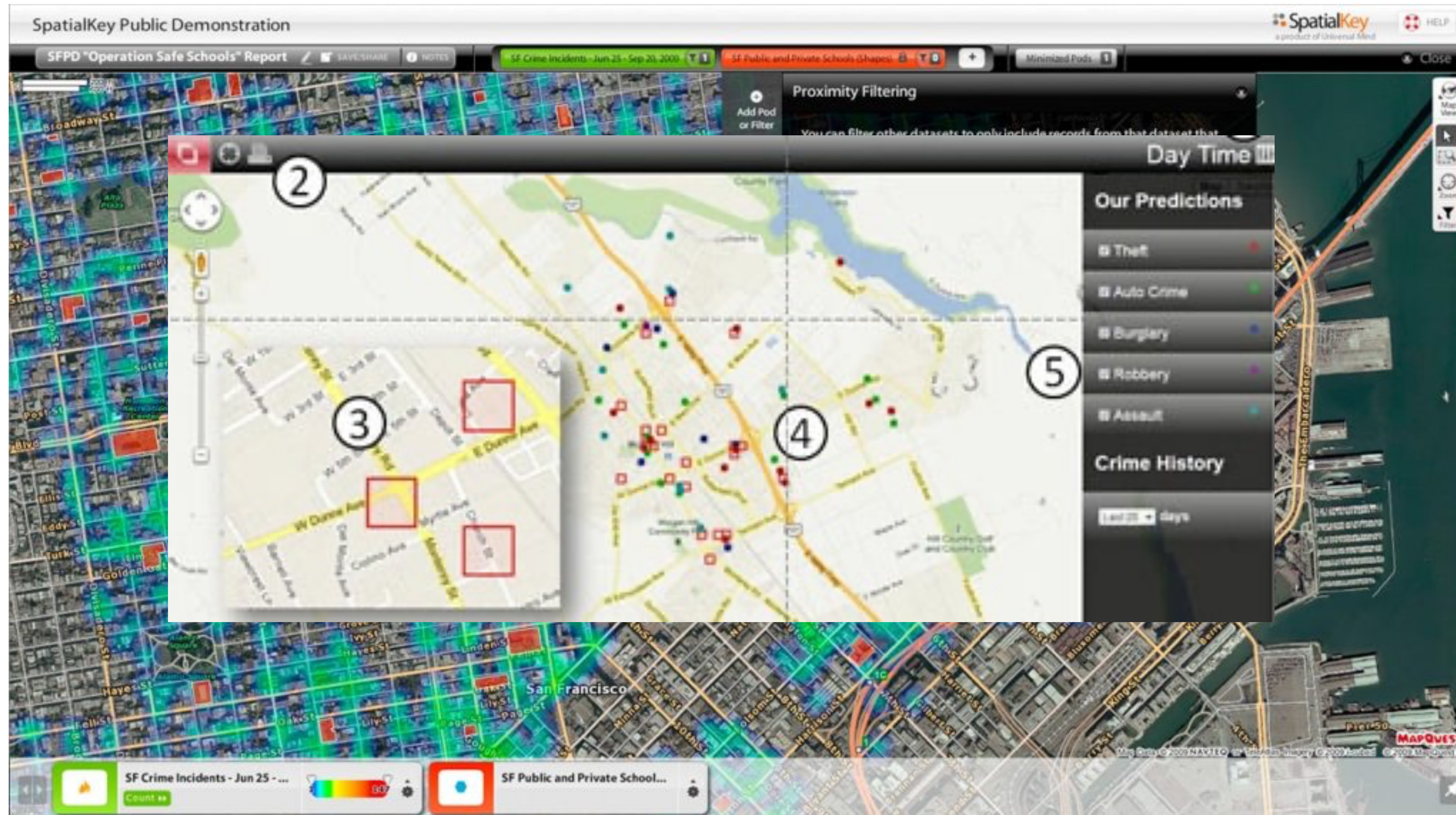
- Bias
- Ethics of use
- Accountability
- Societal impacts
- Economic impacts
- Environmental impacts

Don't be a tool for creating WMDs

- Algorithms (and DS!) implement our biases, yet look objective
- Can implement our biases at scale
- Can have huge impacts on people's lives
- Are not transparent or accountable to the people being impacted

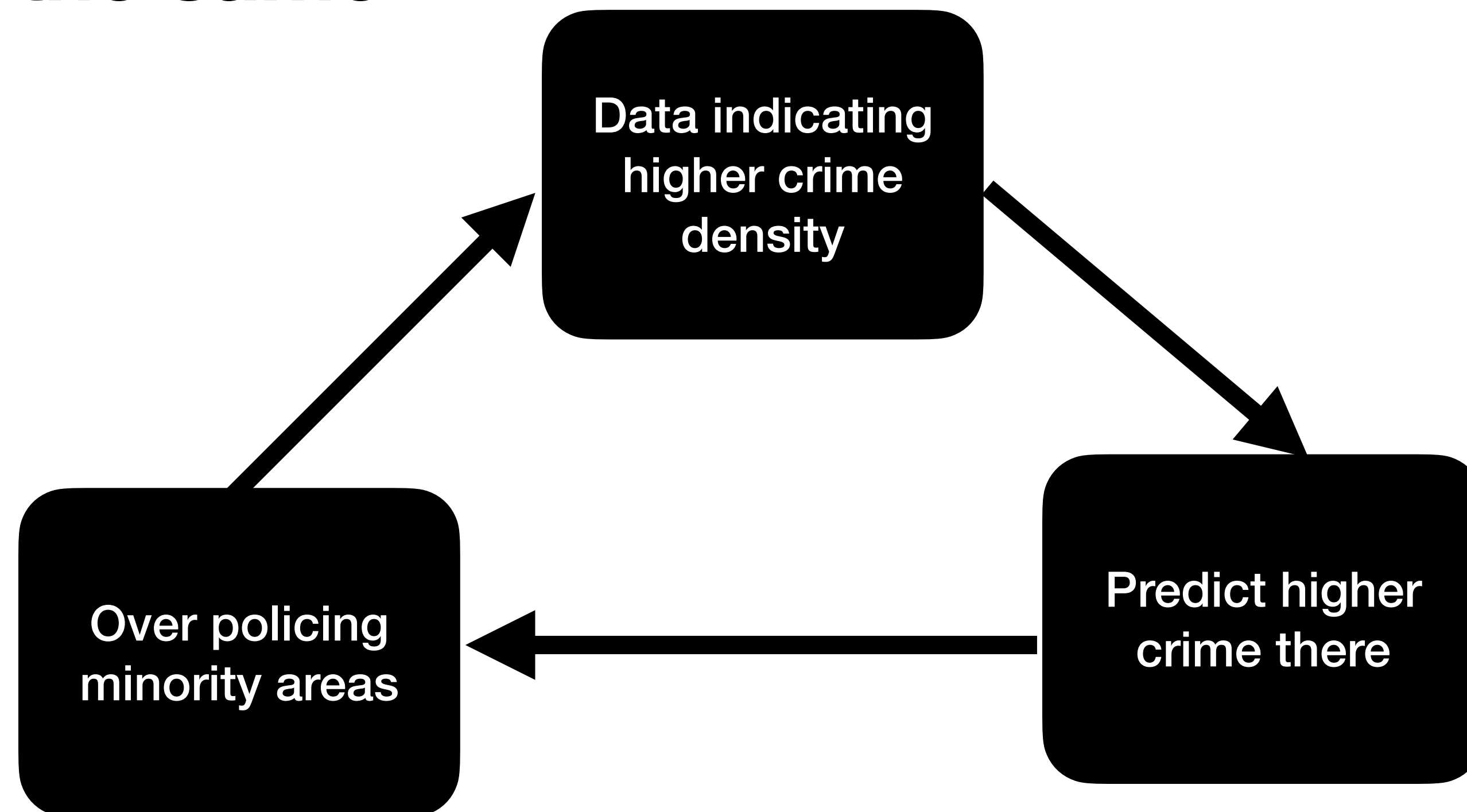


Predictive policing



Predictive policing & sentencing

Blacks arrested for possession at 4x the rate of whites
Usage rates the same





“A lot of times, people are talking about bias in the sense of equalizing performance across groups. They’re not thinking about the underlying foundation, whether a task should exist in the first place, who creates it, who will deploy it on which population, who owns the data, and how is it used?”

-Timnit Gebru

Problems with AI

Same same

- Bias
- Ethics of use
- Accountability
- Societal impacts
- Economic impacts
- Environmental impacts

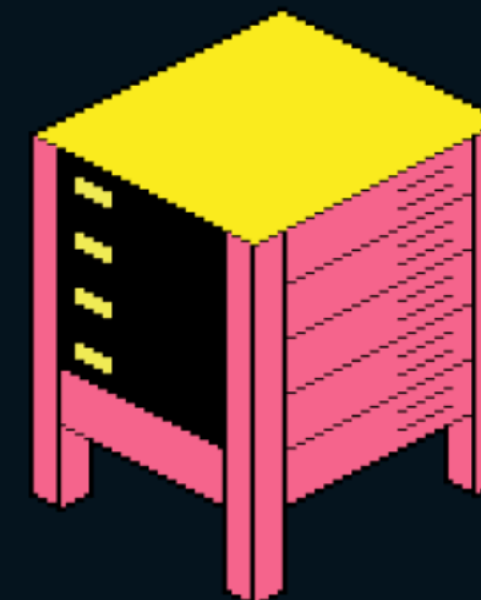
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four years. Google expects to spend \$75 billion on AI infrastructure alone in 2025.

This isn't simply the norm of a digital world. It's unique to AI, and a marked departure from Big Tech's electricity appetite in the recent past. From 2005 to 2017, the amount of electricity going to data centers remained quite flat thanks to increases in efficiency, despite the construction of armies of new data centers to serve the rise of cloud-based online services, from Facebook to Netflix. In 2017, AI began to change everything. Data centers started getting built with energy-intensive hardware designed for AI, which led them to double their electricity consumption by 2023. The latest reports show that 4.4% of all the energy in the US now goes toward data centers.

The carbon intensity of electricity used by data centers was 48% higher than the US average.



Given the direction AI is headed—more personalized, able to reason and solve complex problems on our behalf, and everywhere we look—it's likely that our AI footprint today is the smallest it will ever be. According to new projections published by Lawrence Berkeley National Laboratory in December, by 2028 more than half of the electricity going to data centers will be used for AI. At that point, AI alone could consume as much electricity annually as 22% of all US households.

<https://slate.com/technology/2024/10/uber-lyft-gig-workers-artificial-intelligence-wage-discrimination-jobs.html>

The **San Francisco** Standard

Latest News Politics Business Opinion Culture Food & Drink Sports Podcasts

Business

It’s official. AI is already killing entry-level jobs

A Stanford study finds AI is slashing jobs for 22- to 25-year-olds, with software engineers hit hardest. New grads face a tougher market than ever.

Published Aug. 27, 2025•10:15am

Share

The AI jobpocalypse has begun.

For professions at risk of being replaced by AI, such as software engineering and customer service, entry-level employment has fallen 13% since 2022, a [Stanford study \(opens in new tab\)](#) has found.

The research, which analyzed millions of jobs using data from payroll processing firm ADP, painted an especially grim picture for tech. Employment for software developers ages 22 to 25 has fallen by almost 20% since ChatGPT’s public release.

To hear it from Anthropic CEO Dario Amodei, this is merely the start. AI could wipe out half of entry-level white-collar jobs over the next five years, he [told Axios \(opens in new tab\)](#).

And then walk through this:

<https://www.anthropic.com/research/labor-market-impacts>

SLATE

News & Politics Culture Technology Business Life Advice Podcas

✖ YES... HA HA HA... YES!

THE INDUSTRY

Why You Might Soon Be Paid Like an Uber Driver—Even If You’re Not One

As gig workers’ pay gets slashed by algorithms, experts warn that A.I.-driven wage systems mean that no one’s paycheck is safe.

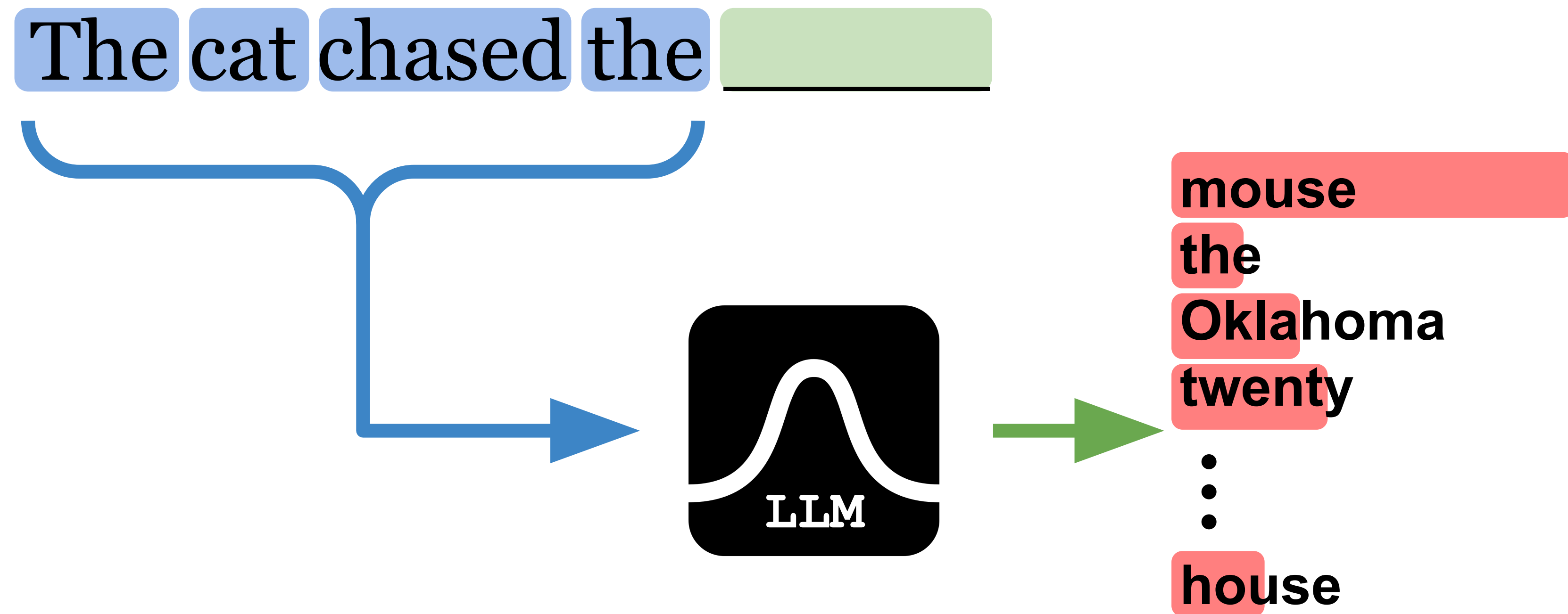
BY MOLLY GLICK OCT 13, 2024 • 5:45 AM

Who owns the stuff?

- AI is a powerful engine of economic inequality
- Copyright cases galore w.r.t training data... fair use or theft?
- Sell a monthly subscription everyone needs to keep their jobs against stiff competition
- This service actually creates that competition
- Instead of lots of jobs, the money (and training data!) goes to AI companies

What is an AI?

AKA Transformer Network, Large Language Model, or Foundation Model



$$P(\text{word}_{n+1} | \text{word}_1, \text{word}_2, \dots, \text{word}_n)$$

Chat ability

<user>Good morning, AI!

</user><AI>Hello there!</AI>

STOP



Good morning, AI!

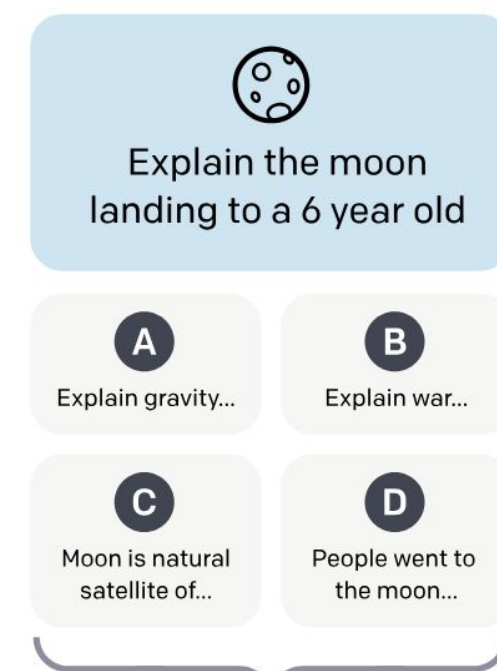
Hello there!

Type here...

Send

Tuning to human preferences

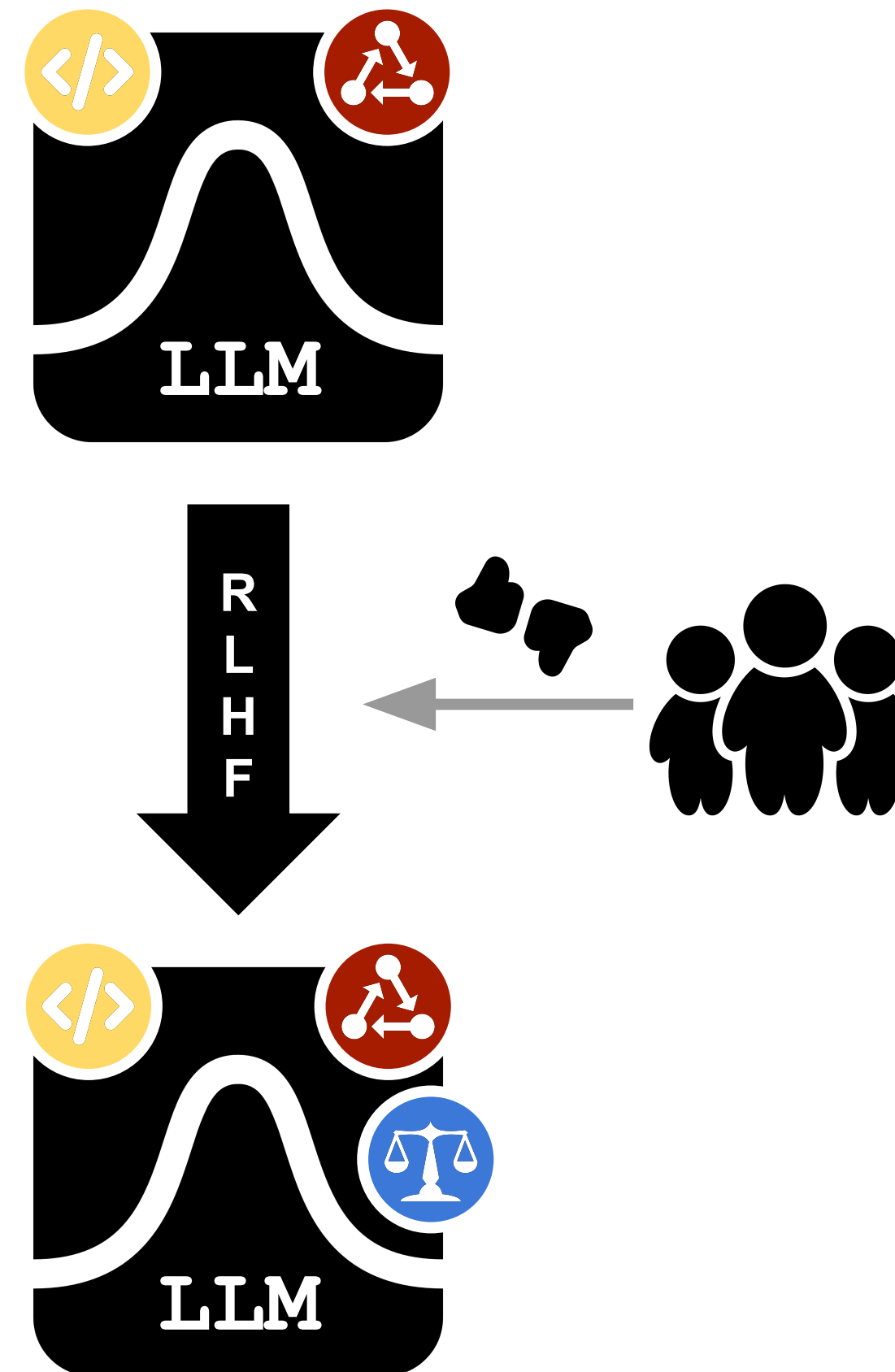
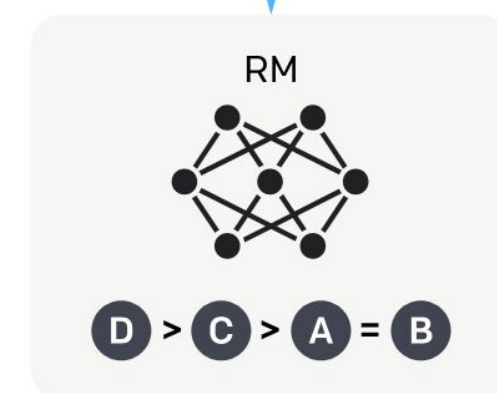
A prompt and several model outputs are sampled.



A labeler ranks the outputs from best to worst.

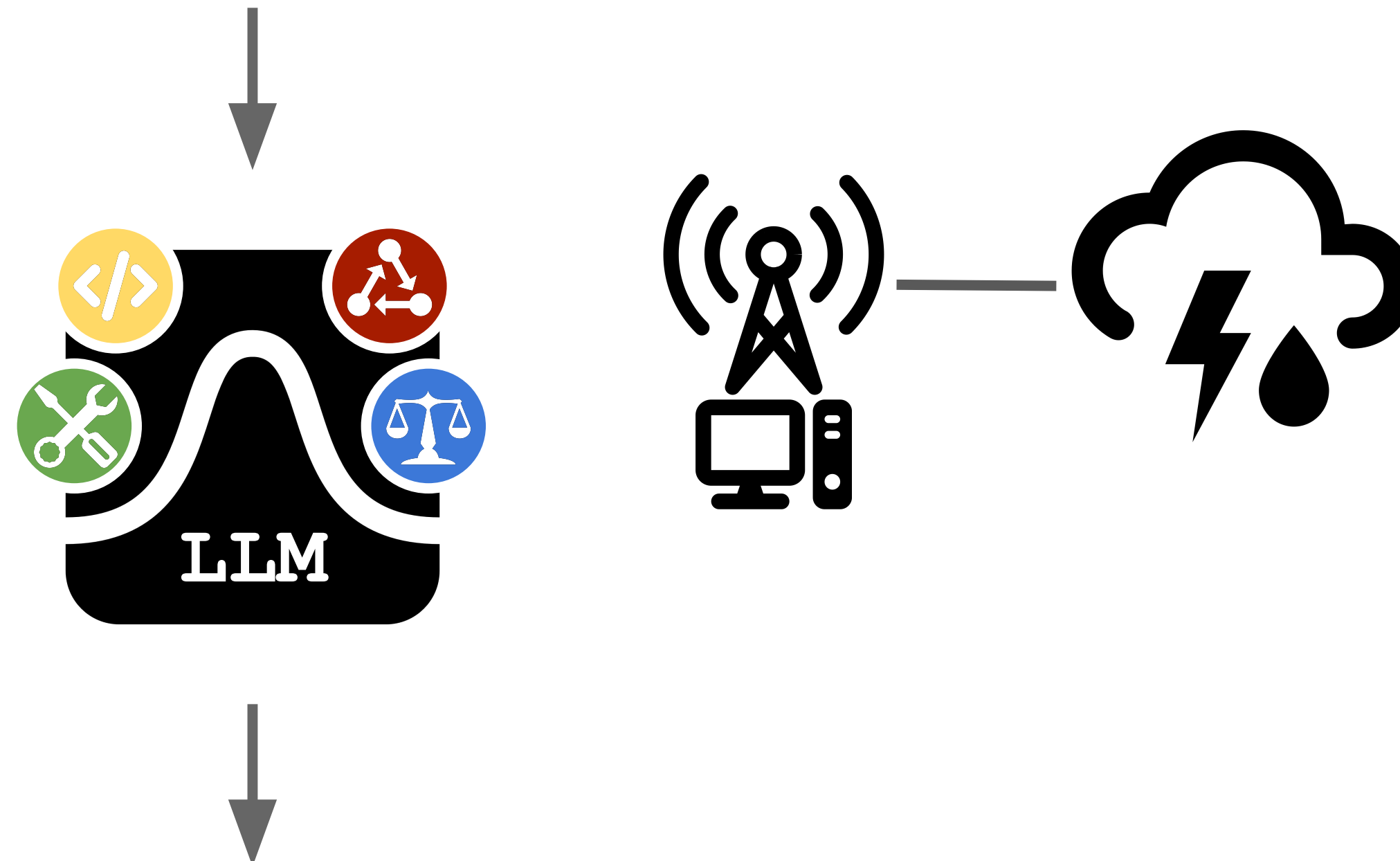


This data is used to train our reward model.



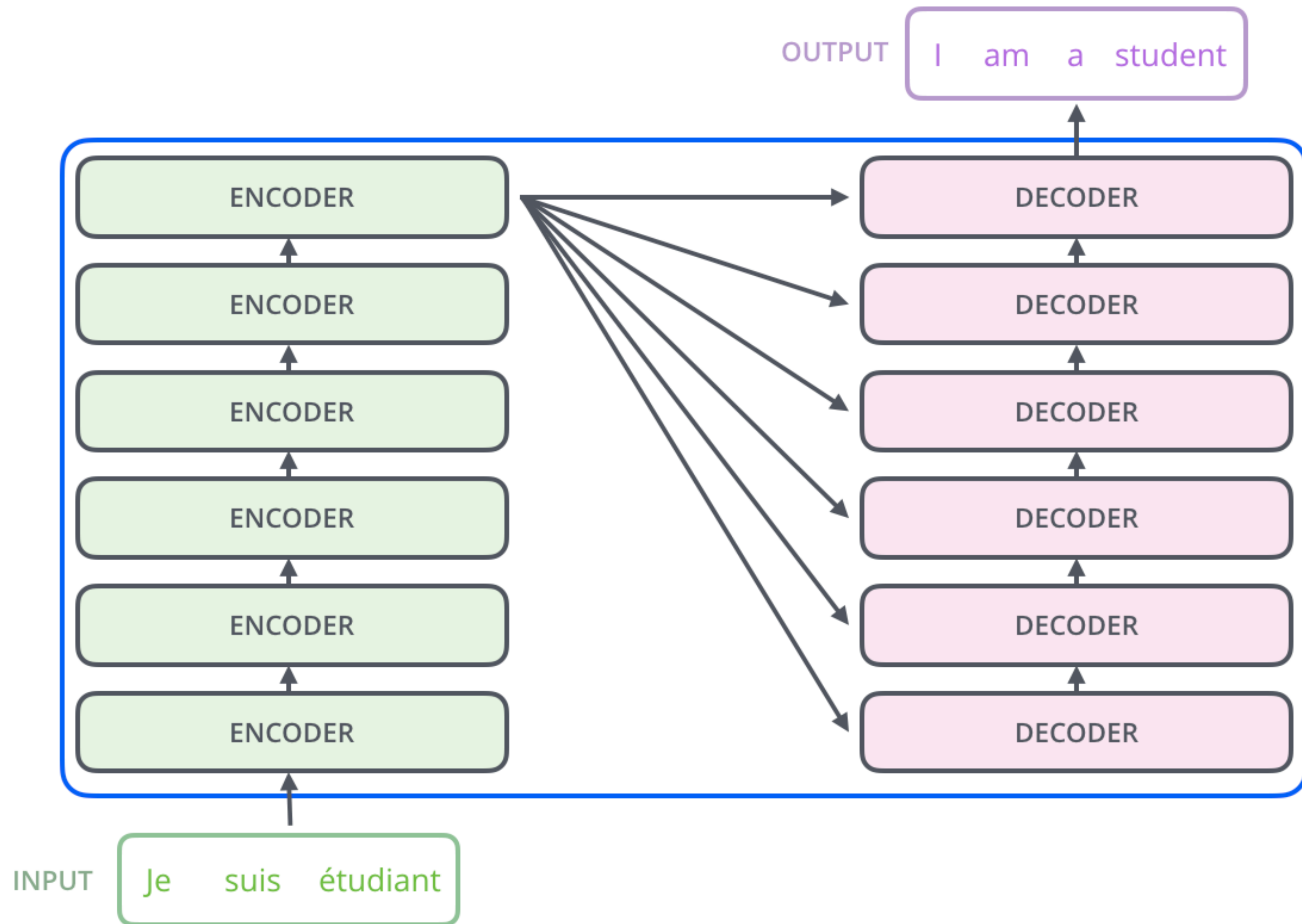
Tool using LLMs

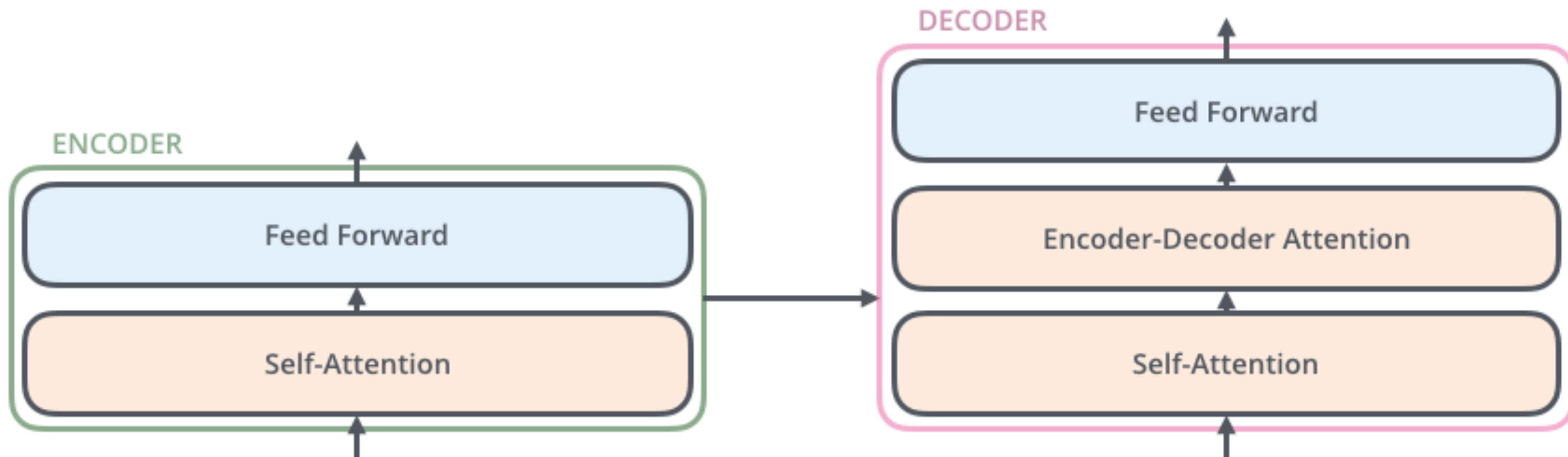
“What is the weather? Will I need an umbrella?”



The weather is currently <tool>get_weather(“san diego”)</tool>







Self-supervised training on large amounts of unlabeled data then small amounts of labelled data for adaptation to a task

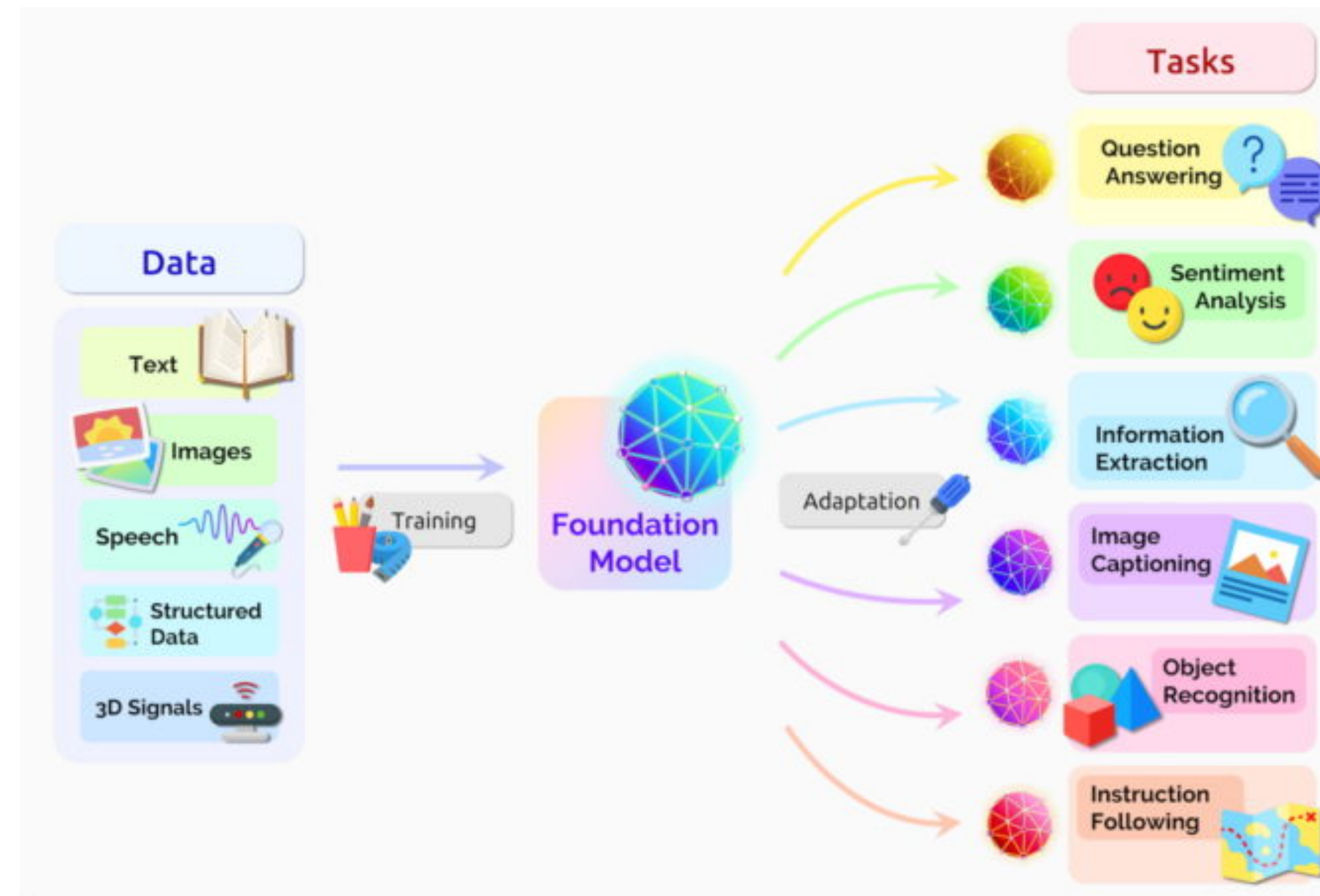
Training:

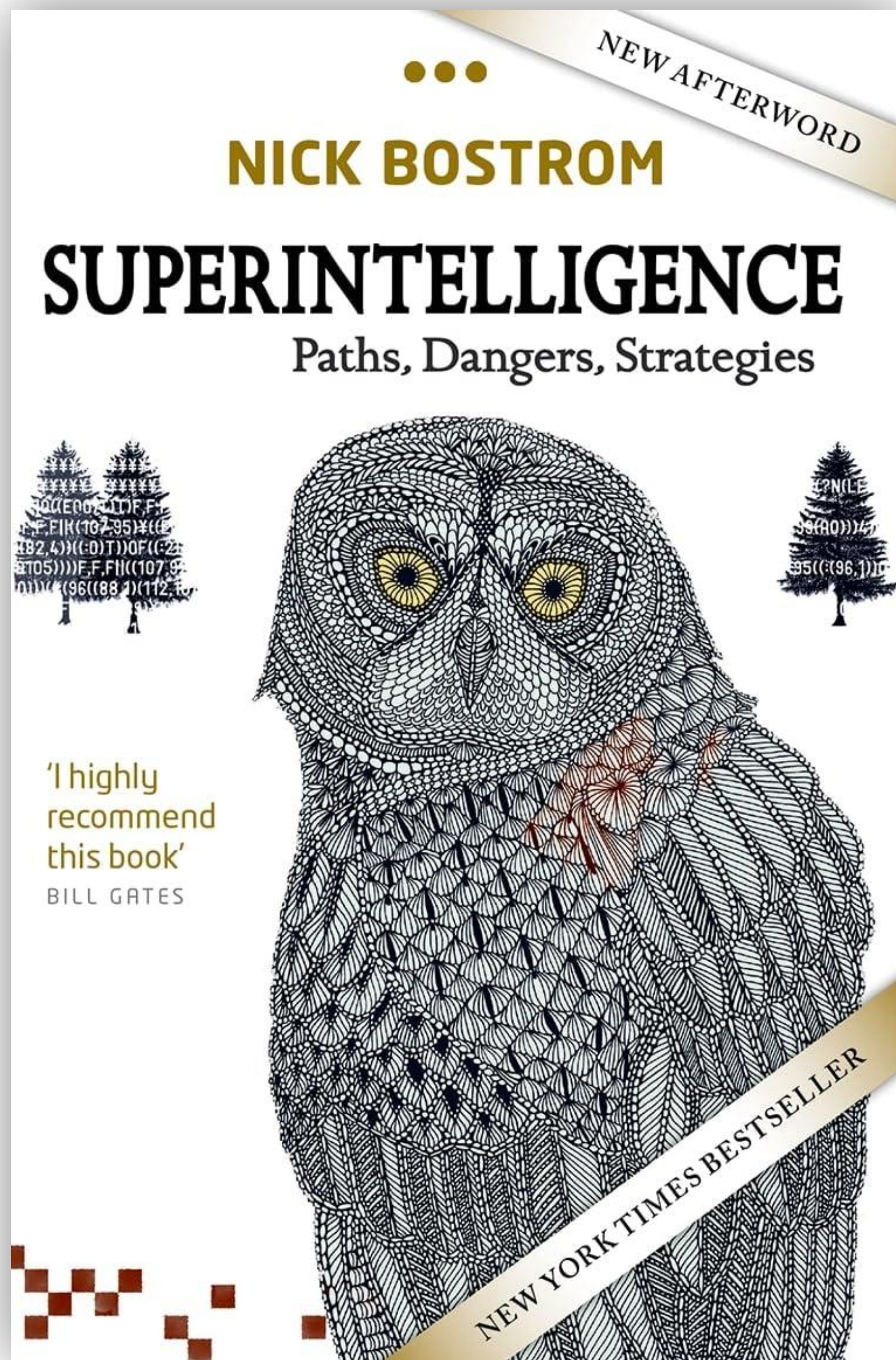
Make LLM predict masked out words
using the rest of the text

the man went to the [MASK] to buy a [MASK] of milk

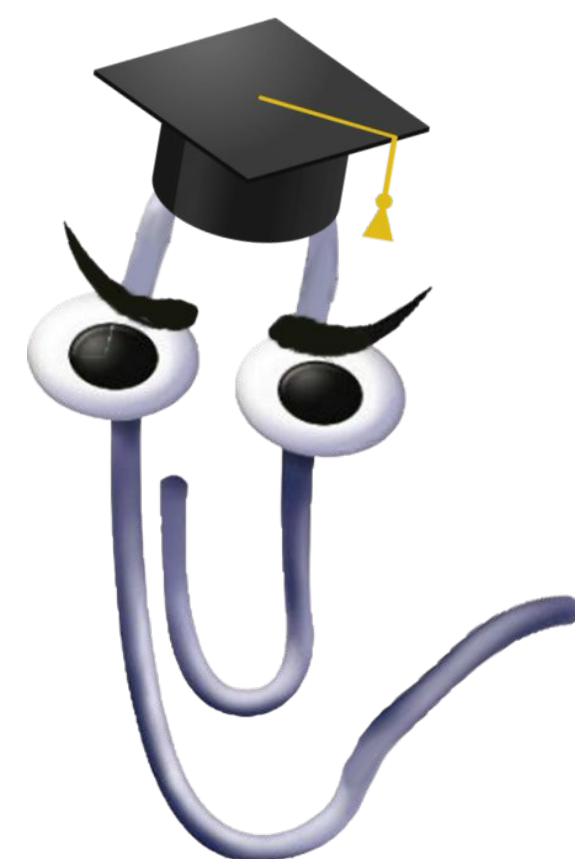
store gallon

↑ ↑

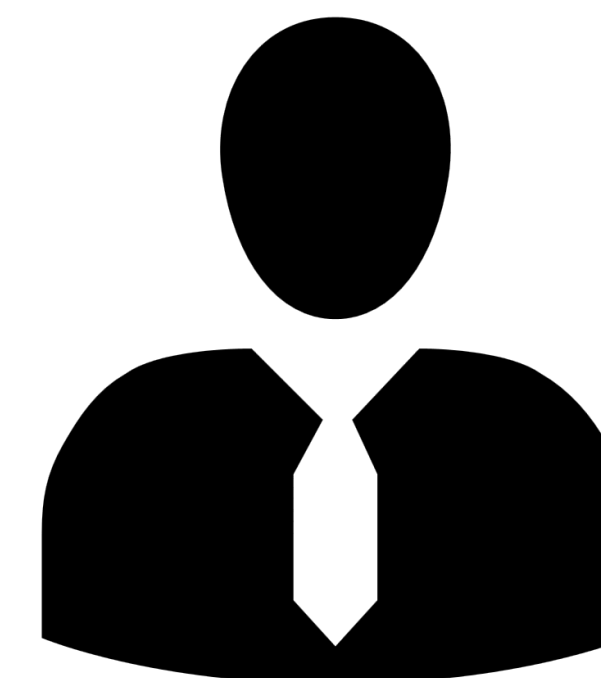




Paperclip AI Thought Experiment

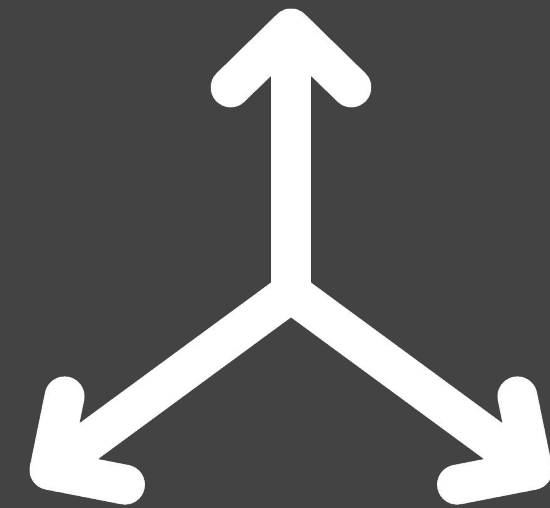


I NEED
LOTS OF
PAPERCLIPS



Orthogonality

Greater intelligence does not imply greater alignment to human values



Instrumental Convergence

All final goals pursued by sufficiently intelligent agents have subgoals in common



(Super)intelligent AI may (un)intentionally make decisions misaligned with human values even while pursuing harmless goals

Cognitive Superpowers

What Bostrom claims an AI would need to be able to do for world domination, human extinction, and other fun activities. What would make an AI a threat?

Task	Skillset
Hacking	Access and manipulate computational resources remotely
Technology Research	Develop technology with real-world possibilities
Economic Productivity	Perform economically productive activity to acquire resources
Strategizing	Achieve distant goals and overcome opposition via planning
Social Manipulation	Leverage human and institutional support via persuasion
Intelligence Amplification	Bootstrap its own capabilities independently

**The alignment problem is fighting
the law of unintended consequences**

LLMs are trained to be polite and not upset users

[Instagram's AI Chatbots Lie About Being Licensed Therapists \(404 media\)](#)

[Most AI Chatbots Easily Tricked Into Giving Dangerous Responses, Study Finds \(the guardian\)](#)

LLMs make it easy to do some kinds of homework

[Ghost Students Are Creating an 'Agonizing' Problem For California Colleges \(sfgate.com\)](#)

Wonder what might go wrong here? Wonder why he wants this?

[Zuckerberg's Grand Vision: Most of Your Friends Will Be AI \(msn.com\)](#)

Broader questions

How can we be more right and less wrong?

Being aware of our biases; external review and checks to minimize

Thinking about tools; know their strengths and limitations

Worry about process and rigorous methods more than outcomes

Social factors influence decision making; good comm and procedures are essential

Be a systems level thinker! Stay curious! Question yourself and your team as much as possible